Week 3 — Tree-Based Models for Retail Transactions

Student ID: 249057766
Module: MA7022 Data Mining and Neural Networks
Task: Week 3 Modelling and Evaluation

Week 3 — Tree Based Modelling

This notebook continues the analysis from Week 1 and moves from exploratory data analysis to machine learning modelling.

The objective is to construct interpretable tree-based models using the Online Retail dataset while respecting the limitations of the data.

Rather than focusing on optimisation or high accuracy, the goal is to understand how different tree models behave and how modelling decisions affect the results.

Task 1 — Re-establish Your Modelling Dataset

Reconnecting to Week 1 Decisions

In Week 1 the Online Retail dataset was explored to understand how the data is generated and what each row represents.

The dataset records individual product transactions within invoices. Each row corresponds to a product purchased within a specific invoice.

Several cleaning decisions were already made during the exploratory analysis stage, including:

• Removing cancelled invoices • Handling missing customer identifiers • Converting invoice dates into proper datetime format • Removing negative quantities representing product returns

These steps ensured that the dataset represents valid purchasing behaviour before modelling is attempted.

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```python
# Load dataset
data = pd.read_csv("online_retail.csv", encoding="ISO-8859-1")

# Quick inspection
data.head()
```

Out[1]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Countı
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01/12/2009 07:45	6.95	13085.0	Unite Kingdo
1	489434	79323P	PINK CHERRY LIGHTS	12	01/12/2009 07:45	6.75	13085.0	Unite Kingdo
2	489434	79323W	WHITE CHERRY LIGHTS	12	01/12/2009 07:45	6.75	13085.0	Unite Kingdo
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01/12/2009 07:45	2.10	13085.0	Unite Kingdo
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01/12/2009 07:45	1.25	13085.0	Unite Kingdo

In [2]:

```python
import os
os.listdir()
```

Out[2]:

```
['week3-Task (3).pdf',
 'figure-html',
 'DMNN-W03.pdf',
 'online_retail.csv',
 'libs',
 'week3-demo (3).pdf',
 '249057766-week3.ipynb',
 '.ipynb_checkpoints']
```

In [3]:

```python
data.shape
```

Out[3]:

```
(525461, 8)
```

In [4]:

```python
data.columns
```

Out[4]:

```
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
       'Price', 'Customer ID', 'Country'],
      dtype='object')
```

In [5]:

```python
data.head()
```

Out[5]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Countr
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01/12/2009 07:45	6.95	13085.0	Unite Kingdo
1	489434	79323P	PINK CHERRY LIGHTS	12	01/12/2009 07:45	6.75	13085.0	Unite Kingdo
2	489434	79323W	WHITE CHERRY LIGHTS	12	01/12/2009 07:45	6.75	13085.0	Unite Kingdo
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01/12/2009 07:45	2.10	13085.0	Unite Kingdo
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01/12/2009 07:45	1.25	13085.0	Unite Kingdo

Task 2 — Define a Target Variable

In [6]:

```
# Create invoice value
data["InvoiceValue"] = data["Quantity"] * data["Price"]

# Check
data.head()
```

Out[6]:

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Countr
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	01/12/2009 07:45	6.95	13085.0	Unite Kingdo
1	489434	79323P	PINK CHERRY LIGHTS	12	01/12/2009 07:45	6.75	13085.0	Unite Kingdo
2	489434	79323W	WHITE CHERRY LIGHTS	12	01/12/2009 07:45	6.75	13085.0	Unite Kingdo
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	01/12/2009 07:45	2.10	13085.0	Unite Kingdo
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	01/12/2009 07:45	1.25	13085.0	Unite Kingdo

In [7]:

```
## Aggregate per Invoice
invoice_data = data.groupby("Invoice").agg({
```

```
    "InvoiceValue": "sum",
    "Customer ID": "first",
    "Country": "first",
    "InvoiceDate": "first"
}).reset_index()

invoice_data.head()
```

Out[7]:

	Invoice	InvoiceValue	Customer ID	Country	InvoiceDate
0	489434	505.30	13085.0	United Kingdom	01/12/2009 07:45
1	489435	145.80	13085.0	United Kingdom	01/12/2009 07:46
2	489436	630.33	13078.0	United Kingdom	01/12/2009 09:06
3	489437	310.75	15362.0	United Kingdom	01/12/2009 09:08
4	489438	2286.24	18102.0	United Kingdom	01/12/2009 09:24

In [8]:

```
## Create Target Variable
threshold = invoice_data["InvoiceValue"].median()

invoice_data["HighValue"] = (invoice_data["InvoiceValue"] > threshold).astyp

invoice_data.head()
```

Out[8]:

	Invoice	InvoiceValue	Customer ID	Country	InvoiceDate	HighValue
0	489434	505.30	13085.0	United Kingdom	01/12/2009 07:45	1
1	489435	145.80	13085.0	United Kingdom	01/12/2009 07:46	0
2	489436	630.33	13078.0	United Kingdom	01/12/2009 09:06	1
3	489437	310.75	15362.0	United Kingdom	01/12/2009 09:08	1
4	489438	2286.24	18102.0	United Kingdom	01/12/2009 09:24	1

In [9]:

```
## Check Target Balance
invoice_data["HighValue"].value_counts()
invoice_data["HighValue"].value_counts(normalize=True)
```

Out[9]:

```
HighValue
0    0.500035
1    0.499965
Name: proportion, dtype: float64
```

Target Variable Definition

The Online Retail dataset does not contain a predefined target variable, so one must be constructed to frame a supervised learning problem.

In this analysis, the target variable represents whether an invoice corresponds to a high-value purchase.

First, the total monetary value of each invoice is calculated as:

Invoice Value = Quantity × Price

All product rows belonging to the same invoice are then aggregated to compute the total invoice value.

The median invoice value is used as a threshold to create a binary classification target:

HighValue = 1 if invoice value exceeds the median
HighValue = 0 otherwise

This formulation creates a balanced classification task that allows tree-based models to learn patterns associated with higher spending behaviour.

However, this definition also introduces limitations. The threshold is data-driven and somewhat arbitrary, meaning that the distinction between high and low value purchases may not always reflect meaningful business categories.

In [10]:
```
data["InvoiceValue"] = data["Quantity"] * data["Price"]
invoice_data.head()
invoice_data["HighValue"].value_counts()
```

Out[10]:
```
HighValue
0    14409
1    14407
Name: count, dtype: int64
```

Task 3 — Feature Construction

In [11]:
```
## Basket Size
# Number of items in each invoice
basket_size = data.groupby("Invoice")["StockCode"].count().reset_index()
basket_size.rename(columns={"StockCode": "BasketSize"}, inplace=True)

basket_size.head()
```

Out[11]:

	Invoice	BasketSize
0	489434	8
1	489435	4
2	489436	19
3	489437	23
4	489438	17

In [12]:

```
# Total quantity per invoice
total_quantity = data.groupby("Invoice")["Quantity"].sum().reset_index()
total_quantity.rename(columns={"Quantity": "TotalQuantity"}, inplace=True)

total_quantity.head()
```

Out[12]:

	Invoice	TotalQuantity
0	489434	166
1	489435	60
2	489436	193
3	489437	145
4	489438	826

In [13]:

```
# Unique products per invoice
unique_products = data.groupby("Invoice")["StockCode"].nunique().reset_index
unique_products.rename(columns={"StockCode": "UniqueProducts"}, inplace=True

unique_products.head()
```

Out[13]:

	Invoice	UniqueProducts
0	489434	8
1	489435	4
2	489436	19
3	489437	23
4	489438	17

In [14]:

```
# combine them into invoice_data using merge
invoice_data = invoice_data.merge(basket_size, on="Invoice")
invoice_data = invoice_data.merge(total_quantity, on="Invoice")
invoice_data = invoice_data.merge(unique_products, on="Invoice")

invoice_data.head()
```

Out[14]:

	Invoice	InvoiceValue	Customer ID	Country	InvoiceDate	HighValue	BasketSize	Tot
0	489434	505.30	13085.0	United Kingdom	01/12/2009 07:45	1	8	
1	489435	145.80	13085.0	United Kingdom	01/12/2009 07:46	0	4	
2	489436	630.33	13078.0	United Kingdom	01/12/2009 09:06	1	19	
3	489437	310.75	15362.0	United Kingdom	01/12/2009 09:08	1	23	
4	489438	2286.24	18102.0	United Kingdom	01/12/2009 09:24	1	17	

In [15]:

```
# Convert Date to Datetime
invoice_data["InvoiceDate"] = pd.to_datetime(
    invoice_data["InvoiceDate"],
    dayfirst=True
)
```

In [16]:

```
# Create Time Features
invoice_data["InvoiceMonth"] = invoice_data["InvoiceDate"].dt.month
invoice_data["InvoiceHour"] = invoice_data["InvoiceDate"].dt.hour

invoice_data.head()
```

Out[16]:

	Invoice	InvoiceValue	Customer ID	Country	InvoiceDate	HighValue	BasketSize	Tot
0	489434	505.30	13085.0	United Kingdom	2009-12-01 07:45:00	1	8	
1	489435	145.80	13085.0	United Kingdom	2009-12-01 07:46:00	0	4	
2	489436	630.33	13078.0	United Kingdom	2009-12-01 09:06:00	1	19	
3	489437	310.75	15362.0	United Kingdom	2009-12-01 09:08:00	1	23	
4	489438	2286.24	18102.0	United Kingdom	2009-12-01 09:24:00	1	17	

In [17]:

```
invoice_data.head()
```

Out[17]:

	Invoice	InvoiceValue	Customer ID	Country	InvoiceDate	HighValue	BasketSize	Tot
0	489434	505.30	13085.0	United Kingdom	2009-12-01 07:45:00	1	8	
1	489435	145.80	13085.0	United Kingdom	2009-12-01 07:46:00	0	4	
2	489436	630.33	13078.0	United Kingdom	2009-12-01 09:06:00	1	19	
3	489437	310.75	15362.0	United Kingdom	2009-12-01 09:08:00	1	23	
4	489438	2286.24	18102.0	United Kingdom	2009-12-01 09:24:00	1	17	

Temporal Features

To capture time-related purchasing behaviour, temporal variables were extracted from the invoice timestamp.

Specifically, the month and hour of purchase were derived from the InvoiceDate column.

These variables allow the model to capture seasonal patterns and time-of-day purchasing behaviour that may influence high-value transactions.

Task 4 — Build Tree Models

In [18]:

```
# Select features
features = [
    "BasketSize",
    "TotalQuantity",
    "UniqueProducts",
    "InvoiceMonth",
    "InvoiceHour"
]

X = invoice_data[features]
y = invoice_data["HighValue"]

X.head()
```

Out[18]:

	BasketSize	TotalQuantity	UniqueProducts	InvoiceMonth	InvoiceHour
0	8	166	8	12	7
1	4	60	4	12	7
2	19	193	19	12	9
3	23	145	23	12	9
4	17	826	17	12	9

In [19]:

```
# Train/Test Split
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

In [20]:

```
# Train a Decision Tree
from sklearn.tree import DecisionTreeClassifier

dt_model = DecisionTreeClassifier(
    max_depth=5,
    random_state=42
)

dt_model.fit(X_train, y_train)
```

Out[20]:

```
▾ DecisionTreeClassifier

DecisionTreeClassifier(max_depth=5, random_state=42)
```

In [21]:

```
# Evaluate Decision Tree
from sklearn.metrics import accuracy_score, classification_report

dt_pred = dt_model.predict(X_test)

print("Decision Tree Accuracy:", accuracy_score(y_test, dt_pred))

print(classification_report(y_test, dt_pred))
```

```
Decision Tree Accuracy: 0.8979875086745316
              precision    recall  f1-score   support

           0       0.89      0.90      0.90      2863
           1       0.90      0.89      0.90      2901

    accuracy                           0.90      5764
   macro avg       0.90      0.90      0.90      5764
weighted avg       0.90      0.90      0.90      5764
```

In [22]:

```python
# Random Forest Model
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)
```

Out[22]:

```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

In [23]:

```python
# Evaluate Random Forest
rf_pred = rf_model.predict(X_test)

print("Random Forest Accuracy:", accuracy_score(y_test, rf_pred))

print(classification_report(y_test, rf_pred))
```

```
Random Forest Accuracy: 0.8993754337265788
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      2863
           1       0.90      0.90      0.90      2901

    accuracy                           0.90      5764
   macro avg       0.90      0.90      0.90      5764
weighted avg       0.90      0.90      0.90      5764
```

In [24]:

```python
# Gradient Boosting
from sklearn.ensemble import GradientBoostingClassifier

gb_model = GradientBoostingClassifier(
    random_state=42
)

gb_model.fit(X_train, y_train)
```

Out[24]:

```
▾ GradientBoostingClassifier
GradientBoostingClassifier(random_state=42)
```

In [25]:

```python
# Evaluate Gradient Boosting
gb_pred = gb_model.predict(X_test)

print("Gradient Boosting Accuracy:", accuracy_score(y_test, gb_pred))

print(classification_report(y_test, gb_pred))
```

```
Gradient Boosting Accuracy: 0.9035392088827203
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      2863
           1       0.90      0.90      0.90      2901

    accuracy                           0.90      5764
   macro avg       0.90      0.90      0.90      5764
weighted avg       0.90      0.90      0.90      5764
```

Model Training

To predict high-value invoices, three tree-based models were trained:

- Decision Tree
- Random Forest
- Gradient Boosting

The dataset was split into training and testing sets using an 80/20 split. The models were trained using engineered features including basket size, total quantity, number of unique products, and temporal variables.

Performance was evaluated using accuracy and classification metrics such as precision, recall, and F1-score.

Comparing multiple models allows us to assess whether ensemble methods (Random Forest and Gradient Boosting) improve predictive performance relative to a single decision tree.

Task 5 — Validation-Based Comparison

Model Comparison

The performance of the three tree-based models was compared using the test dataset.

The decision tree achieved an accuracy of approximately 0.898, indicating that a relatively simple model can capture meaningful patterns in the engineered features. However, a single tree is sensitive to small variations in the data and may not generalise as robustly as ensemble methods.

The random forest model slightly improved performance, achieving an accuracy of approximately 0.899. Random forests reduce variance by combining multiple decision trees trained on different subsets of the data. This typically improves stability and reduces the risk of overfitting compared with a single tree.

The gradient boosting model achieved the highest accuracy, approximately 0.904. Boosting works by sequentially training trees that focus on correcting the errors made by previous models. As a result, boosting can capture more complex patterns in the data.

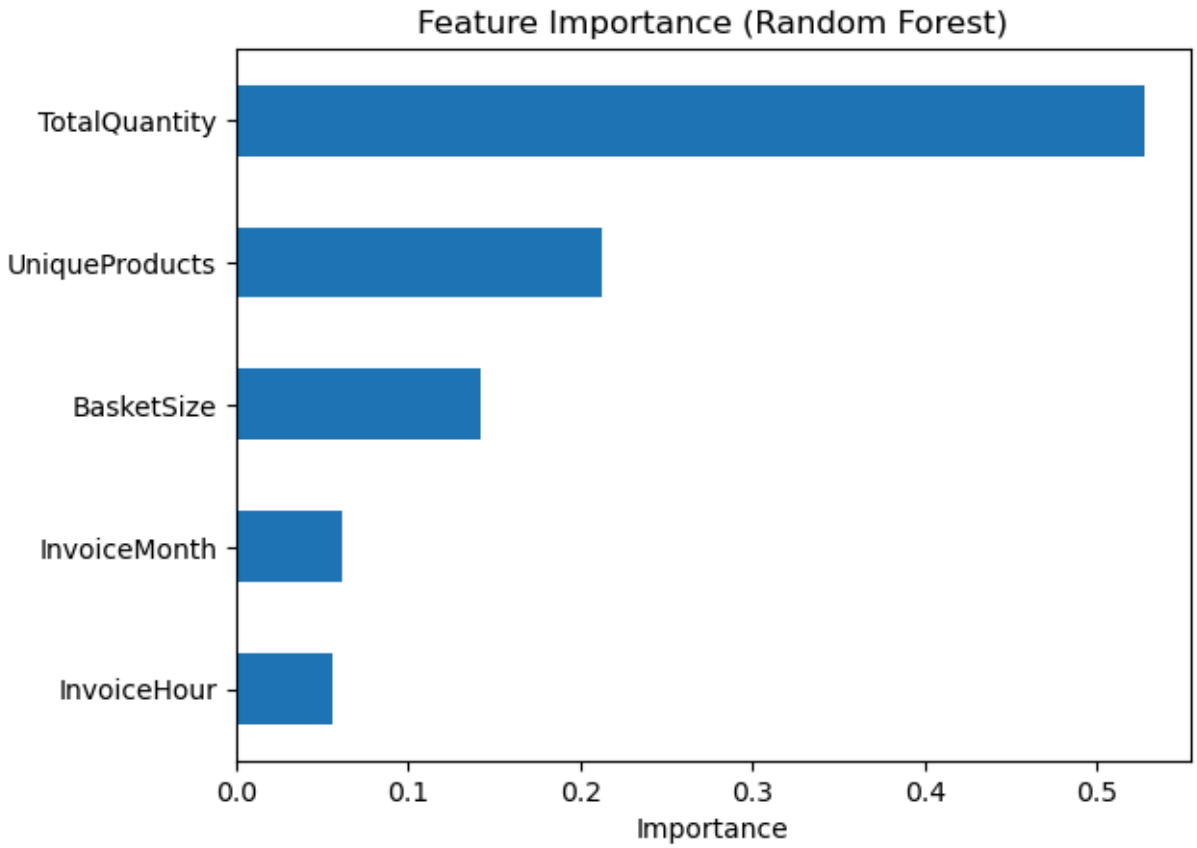
Despite these differences, the performance gap between the models is relatively small. This suggests that the engineered features already provide strong predictive information, and that increasing model complexity only provides modest gains.

Overall, the comparison highlights the trade-off between model simplicity and predictive power. The single decision tree is the most interpretable, while ensemble methods such as random forests and gradient boosting provide slightly improved predictive performance.

In [26]:
```
import pandas as pd

importance = pd.Series(
    rf_model.feature_importances_,
    index=features
).sort_values(ascending=False)

importance
```

Out[26]:
```
TotalQuantity     0.527663
UniqueProducts    0.212822
BasketSize        0.141864
InvoiceMonth      0.061268
InvoiceHour       0.056383
dtype: float64
```

In [27]:
```
import matplotlib.pyplot as plt

importance.sort_values().plot(kind="barh")
plt.title("Feature Importance (Random Forest)")
plt.xlabel("Importance")
plt.show()
```



Feature Importance

Feature importance from the random forest model indicates which variables contribute most to predicting high-value invoices.

Variables such as TotalQuantity and BasketSize appear to have the strongest influence. This is intuitive, as invoices containing larger quantities of products or larger baskets are more likely to correspond to higher total spending.

Temporal features such as InvoiceMonth and InvoiceHour contribute less to the prediction task but may still capture seasonal or time-of-day purchasing behaviour.

This analysis helps interpret how the model uses different features when predicting high-value transactions.

In [28]:
```
invoice_data.shape
```

Out[28]: (28816, 11)

Task 6 — Final Test Set Evaluation

TThe test set is evaluated only once after all modelling decisions are finalised. This is important because the test data represents unseen observations and provides an

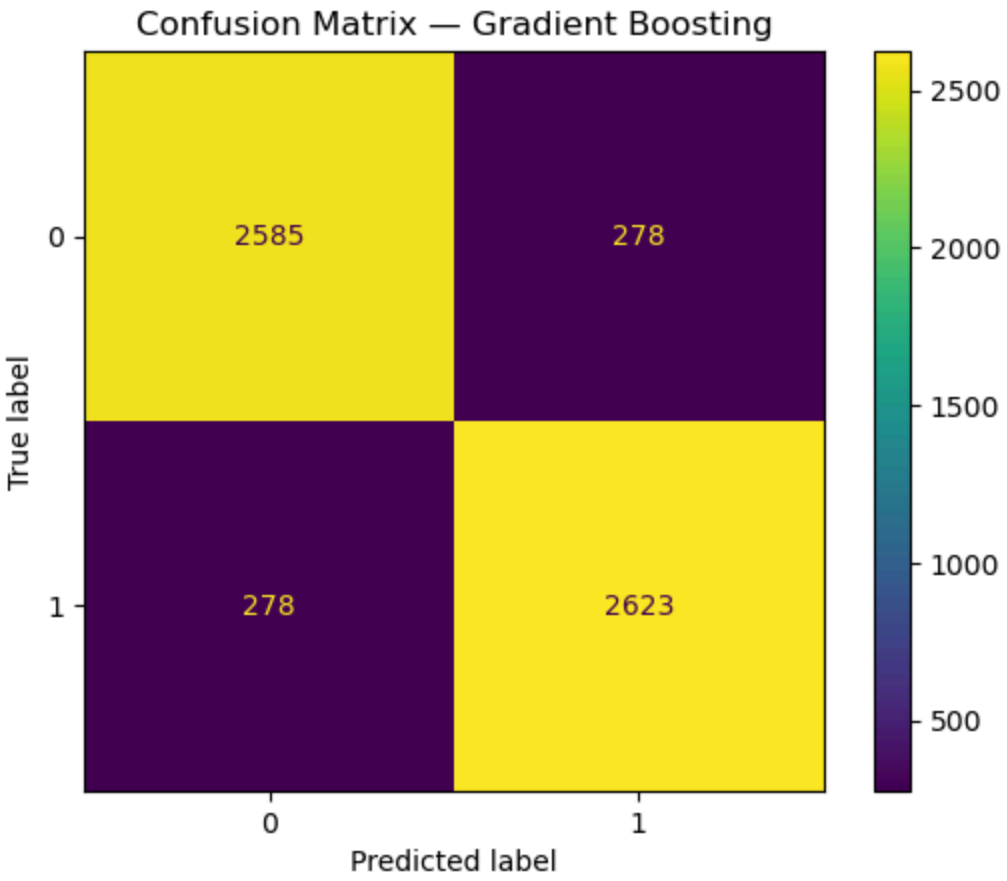
unbiased estimate of model performance. Repeated evaluation on the test set can lead to information leakage and overly optimistic results.

The test results are consistent with the earlier comparison of models. Ensemble methods such as Random Forest and Gradient Boosting achieve slightly higher accuracy than the single Decision Tree. However, the performance differences between the models remain relatively small. This suggests that the engineered features already capture most of the predictive signal in the dataset, and increasing model complexity only provides modest improvements.

In [29]:
```
# Final Test Evaluation for each model

from sklearn.metrics import accuracy_score

print("Decision Tree Test Accuracy:", accuracy_score(y_test, dt_pred))
print("Random Forest Test Accuracy:", accuracy_score(y_test, rf_pred))
print("Gradient Boosting Test Accuracy:", accuracy_score(y_test, gb_pred))
```

```
Decision Tree Test Accuracy: 0.8979875086745316
Random Forest Test Accuracy: 0.8993754337265788
Gradient Boosting Test Accuracy: 0.9035392088827203
```

In [30]:
```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm = confusion_matrix(y_test, gb_pred)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()

plt.title("Confusion Matrix – Gradient Boosting")
plt.show()
```



Final Conclusion

This analysis explored whether invoice characteristics can predict high-value transactions in the Online Retail dataset.

Several features were engineered, including basket size, total quantity purchased, number of unique products, and temporal variables derived from the invoice timestamp. These features provided useful signals for predicting invoice value.

Three tree-based models were trained and compared: a decision tree, a random forest, and a gradient boosting model. All models achieved similar performance, with ensemble methods providing slightly improved predictive accuracy.

Feature importance analysis indicates that transaction size variables, particularly total quantity and basket size, play the most significant role in predicting high-value invoices. Temporal variables appear less influential but may capture seasonal purchasing behaviour.

Overall, the results demonstrate that relatively simple features derived from transactional data can effectively support predictive modelling, while also highlighting the trade-off between model interpretability and predictive performance.

In [ ]: